# GLS versus Averaged Change-Score Estimators for Clinical Trials with Run-In Observations

Ronald G. Thomas, Ph.D.*

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**Abstract**

**Background.** Three estimators of the treatment effect in longitudinal trials with run-in observations are in routine use: the GLS estimator under the linear mixed model, which exploits the full covariance structure to extract maximum information from all observations; the ANCOVA estimator, which regresses post-randomisation means on the run-in mean as a covariate without specifying the random-effects covariance; and the averaged change-score estimator, which computes a simple difference of phase means and applies a $t$-test, requiring no variance-component estimation. The relative efficiency of these estimators under correct and misspecified models has not been comprehensively characterised.

**Methods.** We derive the relative efficiency of the three estimators as a function of the trial design $(J_0, J_1, J_2)$ and the variance parameters $(\sigma^2, \sigma_b^2)$. The derivation proceeds via the Woodbury-based variance formulas of Paper 1 in this compendium, plus closed-form expressions for the ANCOVA and averaged-change-score variances under the same random-slopes model. We complement the analytic comparison with a Monte Carlo simulation that probes robustness to misspecification of the within-subject covariance ($R$) and the random-effects covariance ($G$).

**Results.** Under correct specification, GLS $\leq$ ANCOVA $\leq$ averaged in variance. The efficiency loss from ANCOVA relative to GLS is below 5% across a broad regime of $(J_0, J_1, J_2)$ and signal-to-noise ratios; conditions characterising this regime are derived. Under misspecification of $G$ or $R$, the simpler estimators can outperform GLS: when the variance components are mis-stated, GLS amplifies the misspecification through its inverse- covariance weighting, while ANCOVA and averaged remain comparatively robust.

**Conclusions.** GLS is the optimal estimator under correct model specification but does not dominate the simpler alternatives in the practical regime of mild misspecification. ANCOVA is the recommended default when the variance components are known only approximately; averaged change-score remains useful as a robust benchmark and a teaching tool.

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*Wertheim School of Public Health, UCSD; ORCID: 0000-0003-1686-4965

# 1 Introduction

[1] **The optimality of GLS is contingent on knowing the covariance, which in practice must be estimated and may be unstable in small samples.**

- GLS is the minimum-variance unbiased estimator only when the covariance matrix is known.
- Feasible GLS substitutes estimated variance components, so its efficiency depends on estimation accuracy.
- Small samples or short follow-up yield unstable estimates that can erode the theoretical advantage of GLS.

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*The GLS estimator $\hat{\gamma}_{GLS} = (X'\Sigma^{-1}X)^{-1}X'\Sigma^{-1}\mathbf{c}$ is the minimum-variance unbiased estimator when $\Sigma$ is known. In practice, $\Sigma$ must be estimated, and the efficiency of the feasible GLS depends on the accuracy of this estimation. With small sample sizes or short follow-up, variance component estimates can be unstable, potentially eroding the theoretical advantage of GLS.*

The averaged change-score approach sidesteps variance component estimation entirely. For a trial with $J_0$ run-in and $J_2$ common close observations, the estimator is:

$$\hat{\gamma}_{\text{avg}} = \bar{d}_1 - \bar{d}_0 \tag{1}$$

where

$$d_{ik} = \frac{1}{J_{2,i}+1}\sum_{\ell=0}^{J_{2,i}} Y_{i,q+\ell,k} - \frac{1}{J_{0,i}+1}\sum_{\ell=0}^{J_{0,i}} Y_{i,\ell,k} \tag{2}$$

is the difference of post- and pre-randomization means for participant $i$.

[2] **The relative merits of change-score and ANCOVA analyses have been studied extensively across several methodological traditions.**

- Formal efficiency comparisons for pretest-posttest designs trace back to @brogan1980 and @yang2001.
- Subsequent work contrasted baseline-as-covariate with baseline-as-response and developed semiparametric covariate adjustment.
- Later contributions clarified when longitudinal models gain power over ANCOVA and adapted these ideas to run-in trials.

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*The debate between change-score and ANCOVA analyses has a long history [@senn2006; @vanbreukelen2006; @crager1987]. @yang2001 provided a formal efficiency comparison for pretest-posttest designs. @liu2009 and @lu2010 compared baseline-as-covariate against baseline-as-response (constrained longitudinal data analysis). @tsiatis2008 developed a semiparametric framework for covariate adjustment. @mallinckrodt2021 showed that MMRM requires time-by-covariate interactions to guarantee power gains over*

*ANCOVA. @brogan1980 established the foundational comparison of pretest-posttest designs. @coffman2016 addressed conditioning on baseline in longitudinal RCTs. @wang2019 proposed the two-period approach for run-in trials.*

[3] **The averaged change-score estimator is distinct from and less efficient than the constrained longitudinal data analysis estimator, which is excluded from the present comparison.**

- The cLDA estimator places baseline in the response vector under a common-mean constraint.
- It exploits the full covariance structure without specifying the random-effects distribution, ranking between the averaged estimator and GLS.
- A full comparison including cLDA lies beyond the present scope but constitutes a natural extension.

To be completed by rgt.

*We note that the averaged change-score estimator studied here is distinct from (and less efficient than) the constrained longitudinal data analysis (cLDA) estimator of @liang2000 and @lu2010, which includes baseline in the response vector with a common-mean constraint. The cLDA estimator uses the full covariance structure without requiring specification of the random effects distribution G, placing it between the averaged estimator and GLS in the efficiency ordering. @funatogawa2020 extended the cLDA framework to pre- and post-treatment measurements with equal baseline assumptions. A full comparison including cLDA is beyond the scope of this paper but represents a natural extension.*

[4] **The longpower package implements the GLS baseline but omits the simpler estimators and the misspecification analysis developed here.**

- It is the CRAN reference for slope-comparison power calculations under the Liu-Liang, Diggle, and Edland formulations.
- Its routines correspond to the GLS estimator under correct specification, matching the Paper 1 baseline.
- It does not cover ANCOVA, the averaged change-score test, or the efficiency loss from misspecified covariance.

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*The ‘longpower‘ R package [@iddi2022] provides the CRAN reference implementation of power calculations for the GLS / REML-based slope-comparison estimator, covering the Liu-Liang [@liuliang1997], Diggle [@diggle2002], and Edland [@ard2011] formulations. These routines correspond to the $\hat{\gamma}_{GLS}$ estimator studied here under correct specification and constitute the Paper 1 baseline extended by run-in and common close observations. However, ‘longpower‘ does not cover the ANCOVA-on-run-in-mean estimator or the averaged change-score t-test, nor does it characterize the efficiency loss from misspecification of G or R. The comparison developed here therefore com-*

*plements 'longpower' by (i) ordering the three candidate estimators under correct specification and (ii) quantifying when the simpler estimators outperform GLS under misspecified covariance.*

# 2 Three estimators

## 2.1 GLS under the mixed model

[5] **The GLS variance for the treatment effect follows from the relevant diagonal element of the inverse information matrix under the marginal covariance.**

- The per-pair variance is the treatment-effect element of $(X'\Sigma^{-1}X)^{-1}$.
- The marginal covariance decomposes as residual plus random-effects components.
- The inverse is obtained efficiently via the Woodbury identity.

To be completed by rgt.

*From @frost2008, the per-pair variance is the $(\gamma,\gamma)$ element of $(X'\Sigma^{-1}X)^{-1}$ with $\Sigma = R + ZGZ'$ [@laird1982; @verbeke2000], computed via the Woodbury identity.*

For the Frost conventional design ($J_0 = 0$, $J_1 = 2$):

$$\hat{\gamma}_{\text{GLS}} = \frac{1}{2t}[(C_{22} + C_{21}) - (C_{12} + C_{11})] \tag{3}$$

with $\text{Var}(\hat{\gamma}_{\text{GLS}}) = (\sigma^2 + 2\sigma_b^2 t^2)/t^2$.

## 2.2 Averaged change score ($t$-test)

The averaged change-score estimator ignores the within-subject correlation structure and treats $d_{ik}$ as independent observations:

$$\text{Var}(\hat{\gamma}_{\text{avg}}) = \frac{2}{m}\,\text{Var}(d_{ik}) \tag{4}$$

The variance $\text{Var}(d_{ik})$ depends on the correlation structure but does not require its estimation.

```r
a <- 1.0
b <- 1.0
tt <- 1.0

v_gls <- var_gamma_matrix(2, 2, 0, a, b, tt)
v_avg <- var_gamma_avg(2, 2, 0, a, b, tt)
re <- re_avg_vs_gls(2, 2, 0, a, b, tt)
```

```r
cat(sprintf('J_0=2, J_1=2: GLS=%.4f, Avg=%.4f, RE=%.4f\n',
            v_gls, v_avg, re))
```

```
## J_0=2, J_1=2: GLS=2.1569, Avg=20.0000, RE=0.1078
```

### 2.3 ANCOVA with run-in mean as covariate

The ANCOVA approach uses the averaged run-in rate as a baseline covariate without specifying $\sigma_b^2$:

$$\bar{Y}_{i,\text{post}} = \alpha + \gamma g_i + \phi \bar{Y}_{i,\text{pre}} + \varepsilon_i \tag{5}$$

The variance of $\hat{\gamma}$ under this model is

$$\text{Var}(\hat{\gamma}_{\text{ANC}}) = \frac{2}{m}\,\text{Var}(\bar{Y}_{\text{post}})\,(1 - \rho^2_{\text{pre,post}}) \tag{6}$$

[6] **The ANCOVA variance is governed by the run-in to post-randomization correlation, which determines the magnitude of the variance reduction.**

- The factor $(1 - \rho^2)$ scales the variance by the squared marginal correlation between the phase means.
- With no run-in observations the covariate vanishes and ANCOVA reduces to the averaged estimator.
- A large correlation, arising when the random-slope variance dominates the residual, yields substantial variance reduction.

To be completed by rgt.

*where $\rho_{pre,post}$ is the marginal correlation between the run-in and post-randomization means. When $J_0 = 0$ there is no covariate and ANCOVA reduces to the averaged estimator. When $\rho_{pre,post}$ is large (i.e., when $\sigma_b^2$ dominates $\sigma^2$), the $(1 - \rho^2)$ factor produces substantial variance reduction.*

[7] **ANCOVA occupies a middle ground by estimating the baseline coefficient from the data while still using the predictive run-in observations.**

- Unlike GLS, the coefficient on the run-in mean is estimated empirically rather than derived from assumed variance components.
- Unlike the averaged estimator, it exploits the predictive value of the run-in observations.
- It therefore reduces variance without committing to a full random-effects specification.

To be completed by rgt.

*The key advantage over GLS is that $\phi$ is estimated from the data rather than derived from assumed variance components [@senn2006; @borm2007]. The* *key advantage over the averaged estimator is that it exploits the predictive value of the run-in observations [@vanbreukelen2006; @vickers2001].*

```r
v_gls <- var_gamma_matrix(2, 2, 0, a, b, tt)
v_anc <- var_gamma_ancova(2, 2, 0, a, b, tt)
v_avg <- var_gamma_avg(2, 2, 0, a, b, tt)

cat(sprintf('J_0=2, J_1=2 (sigma_b/sigma = 1):\n'))
```

```
## J_0=2, J_1=2 (sigma_b/sigma = 1):
```

```r
cat(sprintf('  GLS:    %.4f\n', v_gls))
```

```
##   GLS:    2.1569
```

```r
cat(sprintf('  ANCOVA: %.4f\n', v_anc))
```

```
##   ANCOVA: 6.6667
```

```r
cat(sprintf('  Avg:    %.4f\n', v_avg))
```

```
##   Avg:    20.0000
```

```r
cat(sprintf('  RE(GLS/ANC): %.3f\n', v_gls / v_anc))
```

```
##   RE(GLS/ANC): 0.324
```

```r
cat(sprintf('  RE(ANC/Avg): %.3f\n', v_anc / v_avg))
```

```
##   RE(ANC/Avg): 0.333
```

### 2.4 SUR (seemingly unrelated regression)

The SUR approach treats the run-in and treatment periods as separate regression equations linked by correlated errors:

$$Y_1 = X_2 \begin{pmatrix} \beta_{11} \\ \beta_{12} \end{pmatrix} + e_1 \qquad \text{(both groups)} \tag{7}$$

$$Y_2 = X_1 \alpha_2 + e_2 \qquad \text{(baseline)} \tag{8}$$

The deviation-from-mean parameterization gives $\beta_1 = (\delta_1 + \delta_2)/2$ and $\beta_2 = (\delta_2 - \delta_1)/2$, where $\delta_j$ are the group-specific rates.

### 2.5 BLUP as theoretical benchmark

[8] **The three estimators are linear functions of the change scores and admit a common benchmark in the best linear unbiased predictor of the random slopes.**

- Each estimator can be viewed as a linear combination of the observed change scores.

- The BLUP of the subject-specific random slope provides the natural reference predictor.
- It weights the residuals by the inverse marginal covariance using the GLS fixed-effect estimate.

To be completed by rgt.

*The three estimators above are linear functions of the observed change scores and can be compared to a common benchmark: the best linear unbiased predictor (BLUP) of the individual random slopes. Under the random-slopes model with marginal covariance $V_i = R_i + Z_i G Z_i'$, the BLUP of the subject-specific random slope $b_i$ given the data is*

$$\hat{b}_i = G Z_i' V_i^{-1}(y_i - X_i \hat{\beta}) \tag{9}$$

*where $\hat{\beta}$ is the GLS estimator of the fixed effects [@henderson1975; @robinson1991].*

[9] **GLS attains the minimum variance among linear unbiased estimators when the covariance is correct, while the simpler estimators are specific approximations to this benchmark.**

- The group-specific slope estimators inherit the variance of the treatment-effect fixed effect.
- GLS is optimal among linear unbiased estimators under a correctly specified marginal covariance.
- The averaged and ANCOVA estimators are particular linear approximations to this optimal benchmark.

To be completed by rgt.

*The group-specific slope estimators derived from the fixed effects share the same variance as $(\hat{\beta})_\gamma$, and the GLS estimator $\hat{\gamma}_{GLS}$ minimizes the variance among all linear unbiased estimators of $\gamma$ when $V_i$ is correctly specified. The averaged and ANCOVA estimators are specific linear approximations to this benchmark:*

- **Averaged change-score** replaces $V_i^{-1}$ with an unweighted sum; this is optimal only when $V_i \propto I$, which fails for any non-zero random-slope variance.
- **ANCOVA** collapses the run-in visits into a single scalar covariate (the run-in mean) rather than weighting them by $V_i^{-1}$; the resulting efficiency loss vanishes as $J_0 \to \infty$ since the run-in mean becomes a sufficient statistic for the subject-specific intercept plus average slope.

[10] **Under misspecification the simpler approximations can outperform GLS precisely because they rely on fewer variance components.**

- GLS amplifies errors in the assumed covariance through its inverse weighting.
- The averaged and ANCOVA estimators depend on fewer estimated variance components.

- This reduced dependence confers robustness when the model is misspecified.

To be completed by rgt.

*Under model misspecification (Section 4), these approximations can outperform the nominally optimal GLS because they depend on fewer variance components.*

# 3 Relative efficiency under correct specification

## 3.1 Analytical comparison

The relative efficiency of the averaged estimator to GLS is:

$$\mathrm{RE}_{\mathrm{avg}} = \frac{\mathrm{Var}(\hat{\gamma}_{\mathrm{GLS}})}{\mathrm{Var}(\hat{\gamma}_{\mathrm{avg}})} \leq 1 \tag{10}$$

with equality when the averaged estimator happens to coincide with GLS (which occurs in special cases).

Table 1: RE of averaged change-score vs GLS ($\mathrm{Var}_{\mathrm{GLS}} / \mathrm{Var}_{\mathrm{avg}}$) by $\sigma_b/\sigma$.

| $(J_0, J_1, J_2)$ | 0.5 | 1 | 2 | 5 | 20 |
|---|---|---|---|---|---|
| (1,1,0) | 0.667 | 0.500 | 0.250 | 0.056 | 0.004 |
| (2,1,0) | 0.653 | 0.344 | 0.117 | 0.021 | 0.001 |
| (4,1,0) | 0.427 | 0.149 | 0.041 | 0.007 | 0.000 |
| (0,2,0) | 0.364 | 0.400 | 0.429 | 0.442 | 0.444 |
| (1,2,0) | 0.235 | 0.182 | 0.091 | 0.020 | 0.001 |
| (2,2,0) | 0.218 | 0.108 | 0.036 | 0.006 | 0.000 |
| (4,2,0) | 0.123 | 0.040 | 0.011 | 0.002 | 0.000 |
| (0,3,0) | 0.193 | 0.225 | 0.242 | 0.249 | 0.250 |
| (1,3,0) | 0.125 | 0.099 | 0.048 | 0.010 | 0.001 |
| (2,3,0) | 0.108 | 0.052 | 0.017 | 0.003 | 0.000 |
| (4,3,0) | 0.055 | 0.017 | 0.005 | 0.001 | 0.000 |
| (0,1,1) | 0.727 | 0.533 | 0.245 | 0.050 | 0.003 |
| (1,1,1) | 0.452 | 0.258 | 0.093 | 0.017 | 0.001 |
| (2,1,1) | 0.422 | 0.181 | 0.055 | 0.009 | 0.001 |
| (4,1,1) | 0.255 | 0.080 | 0.021 | 0.003 | 0.000 |
| (0,2,1) | 0.254 | 0.186 | 0.081 | 0.016 | 0.001 |
| (1,2,1) | 0.161 | 0.094 | 0.034 | 0.006 | 0.000 |
| (2,2,1) | 0.144 | 0.062 | 0.019 | 0.003 | 0.000 |
| (4,2,1) | 0.079 | 0.025 | 0.007 | 0.001 | 0.000 |
| (0,3,1) | 0.135 | 0.096 | 0.040 | 0.008 | 0.000 |
| (1,3,1) | 0.088 | 0.051 | 0.018 | 0.003 | 0.000 |
| (2,3,1) | 0.075 | 0.032 | 0.010 | 0.002 | 0.000 |

| $(J_0, J_1, J_2)$ | 0.5 | 1 | 2 | 5 | 20 |
|---|---|---|---|---|---|
| (4,3,1) | 0.038 | 0.012 | 0.003 | 0.001 | 0.000 |

Table 2: RE of ANCOVA vs GLS ($\mathrm{Var}_{\mathrm{GLS}}$ / $\mathrm{Var}_{\mathrm{ANC}}$) by $\sigma_b/\sigma$.

| $(J_0, J_1, J_2)$ | 0.5 | 1 | 2 | 5 | 20 |
|---|---|---|---|---|---|
| (1,1,0) | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| (2,1,0) | 0.971 | 0.909 | 0.860 | 0.838 | 0.834 |
| (4,1,0) | 0.838 | 0.747 | 0.713 | 0.702 | 0.700 |
| (0,2,0) | 0.364 | 0.400 | 0.429 | 0.442 | 0.444 |
| (1,2,0) | 0.381 | 0.385 | 0.379 | 0.372 | 0.371 |
| (2,2,0) | 0.359 | 0.324 | 0.298 | 0.288 | 0.286 |
| (4,2,0) | 0.283 | 0.242 | 0.228 | 0.223 | 0.222 |
| (0,3,0) | 0.193 | 0.225 | 0.242 | 0.249 | 0.250 |
| (1,3,0) | 0.202 | 0.204 | 0.196 | 0.189 | 0.188 |
| (2,3,0) | 0.183 | 0.161 | 0.145 | 0.139 | 0.138 |
| (4,3,0) | 0.134 | 0.112 | 0.105 | 0.103 | 0.102 |
| (0,1,1) | 0.727 | 0.533 | 0.245 | 0.050 | 0.003 |
| (1,1,1) | 0.733 | 0.545 | 0.390 | 0.319 | 0.304 |
| (2,1,1) | 0.697 | 0.542 | 0.460 | 0.430 | 0.425 |
| (4,1,1) | 0.587 | 0.490 | 0.456 | 0.446 | 0.445 |
| (0,2,1) | 0.254 | 0.186 | 0.081 | 0.016 | 0.001 |
| (1,2,1) | 0.260 | 0.195 | 0.138 | 0.113 | 0.108 |
| (2,2,1) | 0.244 | 0.190 | 0.161 | 0.150 | 0.148 |
| (4,2,1) | 0.194 | 0.160 | 0.149 | 0.145 | 0.145 |
| (0,3,1) | 0.135 | 0.096 | 0.040 | 0.008 | 0.000 |
| (1,3,1) | 0.138 | 0.102 | 0.071 | 0.058 | 0.055 |
| (2,3,1) | 0.127 | 0.098 | 0.082 | 0.077 | 0.076 |
| (4,3,1) | 0.096 | 0.078 | 0.073 | 0.071 | 0.071 |

### 3.2 When does the gap exceed 5%?

## 4 Robustness to misspecification

### 4.1 Misspecified $G$

[11] **Misestimating the random-slope variance forfeits the optimality of GLS while leaving the averaged estimator unchanged.**

- An incorrect random-slope variance renders the feasible GLS no longer minimum-variance.
- The error enters through the inverse-covariance weighting of GLS.
- The averaged change-score estimator does not use this variance component and is unaffected.

To be completed by rgt.

*When $\sigma_b^2$ is misestimated (e.g., too large or too small), the feasible GLS is no longer optimal. The averaged change-score estimator is unaffected.*

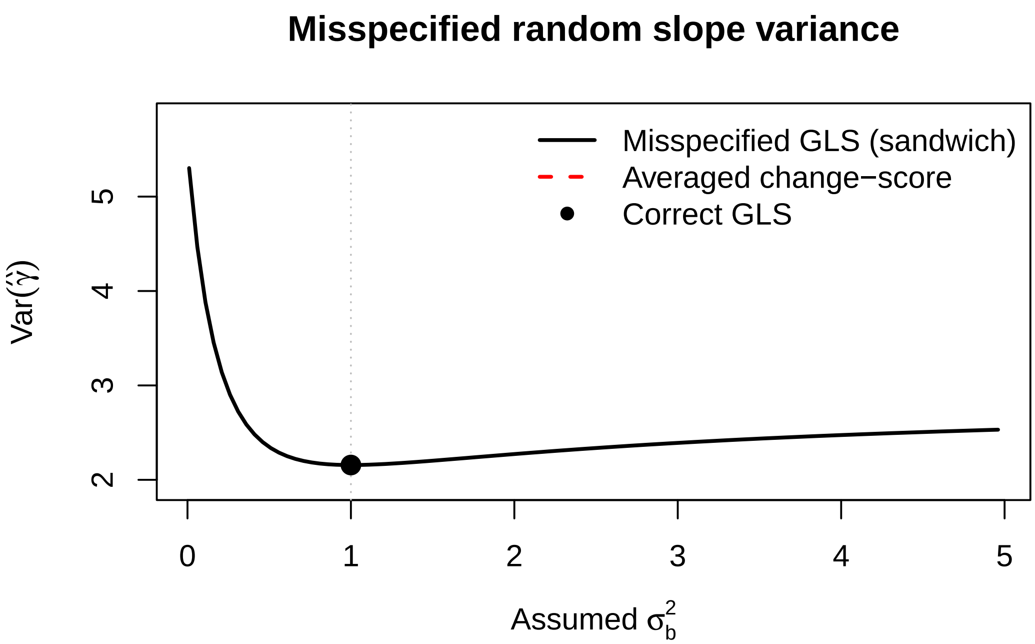


Figure 1: Effect of misspecifying $\sigma_b^2$ on GLS variance. True $\sigma_b^2 = 1$. The GLS estimator uses an assumed value on the x-axis. The averaged estimator (horizontal line) is unaffected.

### 4.2 Misspecified $R$

[12] **Misspecifying the residual correlation biases the GLS variance machinery but again leaves the averaged estimator intact.**

- Assuming compound symmetry when the truth is AR(1) is a representative misspecification.
- Both the GLS variance formula and the Woodbury correction become biased.
- The averaged estimator does not depend on the assumed correlation and is unaffected.

To be completed by rgt.

*When the residual correlation structure is wrong (e.g., compound symmetry assumed but AR(1) is true), both the GLS variance formula and the Woodbury correction are biased. The averaged estimator is again unaffected.*

## 5 Simulation study

### 5.1 Design

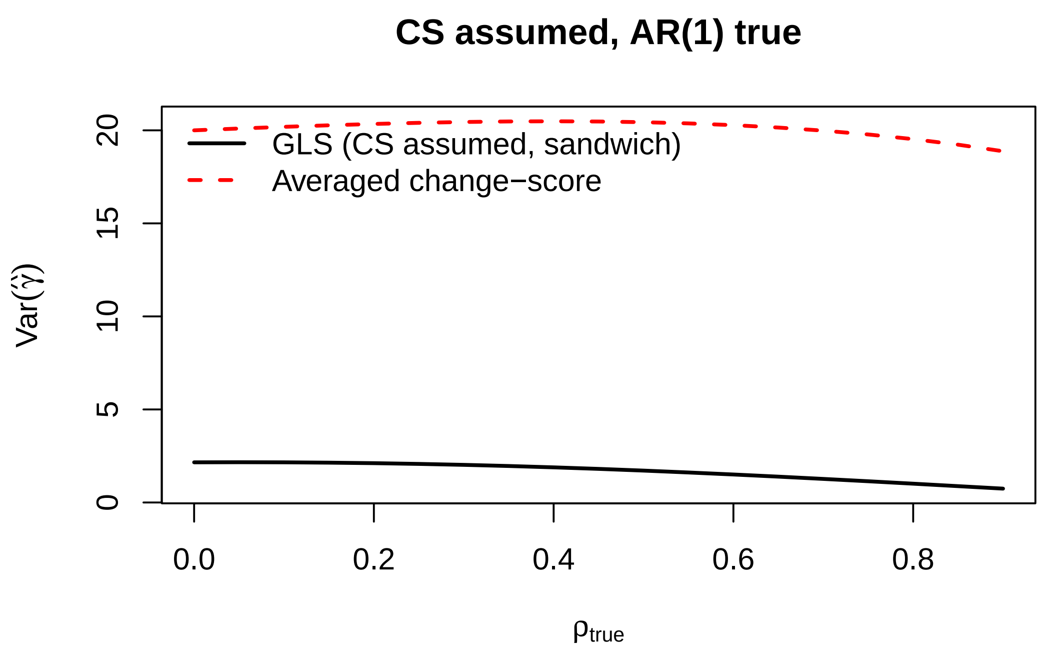


Figure 2: Effect of analyzing under compound symmetry when the true correlation is AR(1). Sandwich variance of GLS (solid) vs averaged change-score (dashed) as a function of $\rho$.

[13] **A factorial Monte Carlo design complements the analytical results by varying sample size, design, signal-to-noise, correlation, and analysis method.**

- Sample size and the run-in design configuration are crossed across several levels.
- The signal-to-noise ratio and the true correlation structure are varied.
- Three analysis methods are compared: correctly specified GLS, misspecified GLS, and the averaged change-score test.

To be completed by rgt.

*To complement the analytical results, we simulate trials under the following factorial design:*

- $n \in \{30, 50, 100, 200\}$ per group
- $(J_0, J_1, J_2) \in \{(0, 2, 0), (2, 2, 0), (2, 2, 2)\}$
- $\sigma_b/\sigma \in \{1, 5, 20\}$
- True correlation: compound symmetry or AR(1)
- Analysis: GLS (correctly specified), GLS (misspecified), averaged change-score

[14] **Each configuration is replicated extensively and evaluated on variance, power, and interval coverage.**

- Five thousand trials are simulated per configuration.
- Empirical variance and empirical power are recorded for each

estimator.

- Coverage of nominal 95% confidence intervals is assessed.

To be completed by rgt.

*For each configuration, 5000 trials are simulated. The outcomes are empirical variance, empirical power, and coverage of nominal 95% confidence intervals.*

```
## Single trial (m=100, delta=0.5):
##   GLS: est=0.694, se=0.147
##   Avg: est=1.154, se=0.432
```

# 6 Numerical examples

## 6.1 MIRIAD parameters

Table 3: Sample size per group: GLS, ANCOVA, and averaged estimators (MIRIAD parameters, $\Delta = 0.25$, 90% power).

| $(J_0, J_1, J_2)$ | $N$ GLS | $N$ ANC | $N$ Avg |
|---|---|---|---|
| (4,3,1) | 36 | 480 | 8249 |
| (4,3,0) | 43 | 399 | 6702 |
| (4,2,1) | 60 | 399 | 6702 |
| (2,3,1) | 69 | 817 | 5328 |
| (4,2,0) | 78 | 339 | 5328 |
| (2,3,0) | 93 | 627 | 4109 |
| (1,3,1) | 99 | 1290 | 4156 |
| (2,2,1) | 104 | 627 | 4109 |
| (0,3,1) | 112 | 2190 | 2190 |
| (1,2,1) | 138 | 931 | 3100 |
| (2,2,0) | 145 | 478 | 3063 |
| (0,2,1) | 150 | 1462 | 1462 |
| (4,1,1) | 156 | 339 | 5328 |
| (1,3,0) | 185 | 931 | 3100 |
| (2,1,1) | 226 | 478 | 3063 |

## 6.2 When to use which estimator

# 7 Discussion

# 8 Conflict of interest

The author declares no conflicts of interest.

9 Funding

This work received no specific funding.

## 10 Data availability statement

[15] **The package implementing all three estimators, the simulation drivers, and the shared bibliography are openly available in the project repository.**

- The estimator implementations reside in the package source files.
- The simulation drivers and shared compendium bibliography are included.
- Specific paths appear in the Reproducibility section below.

To be completed by rgt.

*The ‘runinpower‘ R package implementing the GLS, ANCOVA, and averaged-change-score estimators (‘ancova.R‘, ‘averaged.R‘), the simulation drivers, and the shared bibliography across the four-paper compendium are openly available in the project repository. Specific paths are listed in the Reproducibility section below.*

## 11 Reproducibility

[16] **The manuscript was rendered with R 4.4.x using the package source and a small set of standard analysis and build packages.**

- The in-package source supplies the estimator implementations.
- The GLS comparator in the simulation relies on nlme, with tinytest as the testing harness.
- The manuscript is built with rmarkdown and bookdown.

To be completed by rgt.

*This manuscript was rendered with R 4.4.x. Principal packages used are the in-package ‘runinpower‘ source, ‘nlme‘ (the GLS fitter for the simulation comparator), ‘tinytest‘ (testing harness), and ‘rmarkdown‘ plus ‘bookdown‘ for the manuscript build.*

[17] **The simulation uses a fixed master seed with deterministic per-replicate offsets to ensure reproducibility.**

- A single master seed is set at the entry point of the Monte Carlo comparison.
- Per-replicate seeds derive deterministically from the master seed by adding the replicate index.
- This scheme makes the full simulation exactly reproducible.

To be completed by rgt.